10/14/2026: Induction

CSCI 246: Discrete Structures

Textbook reference: Scheinerman Sec. 22

Midterm Pickup

Midterms are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The midterms are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Motivation: This course and the current job market

- **Mathematical reasoning is harder for AI to replace.** The work most exposed to automation is routine junior coding. Understanding why a model or algorithm behaves the way it does, judging whether an experiment is sound and reasoning about uncertainty are harder to automate.
- **Specialized roles are doing better than generalist ones**: For example, AI/ML engineering is up 39% since 2022.

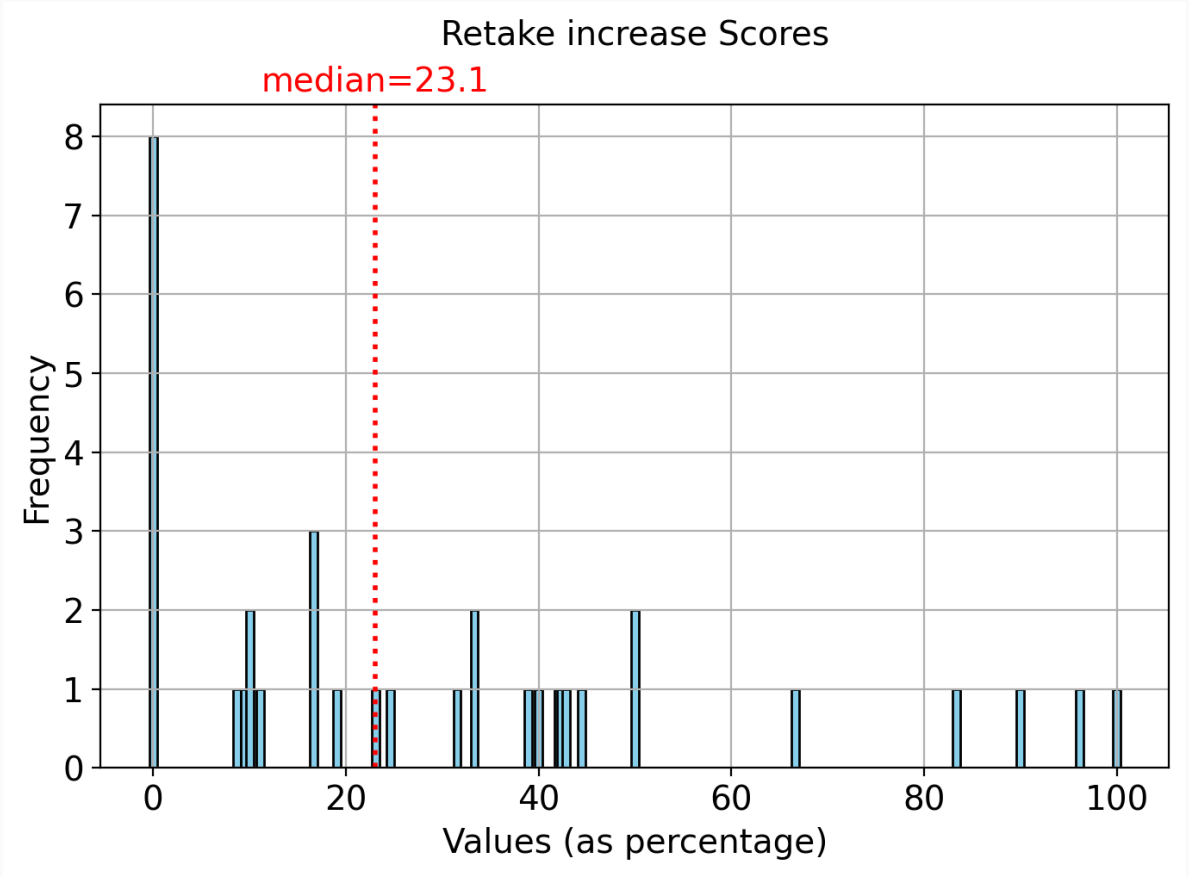
Today's Agenda

- Quiz retake, midterm Review ($\approx$ 5 mins)
- Mini-lecture ($\approx$ 25 mins)
- Group exercises ($\approx$ 20 mins)

Notes on Quiz Retake, Midterm

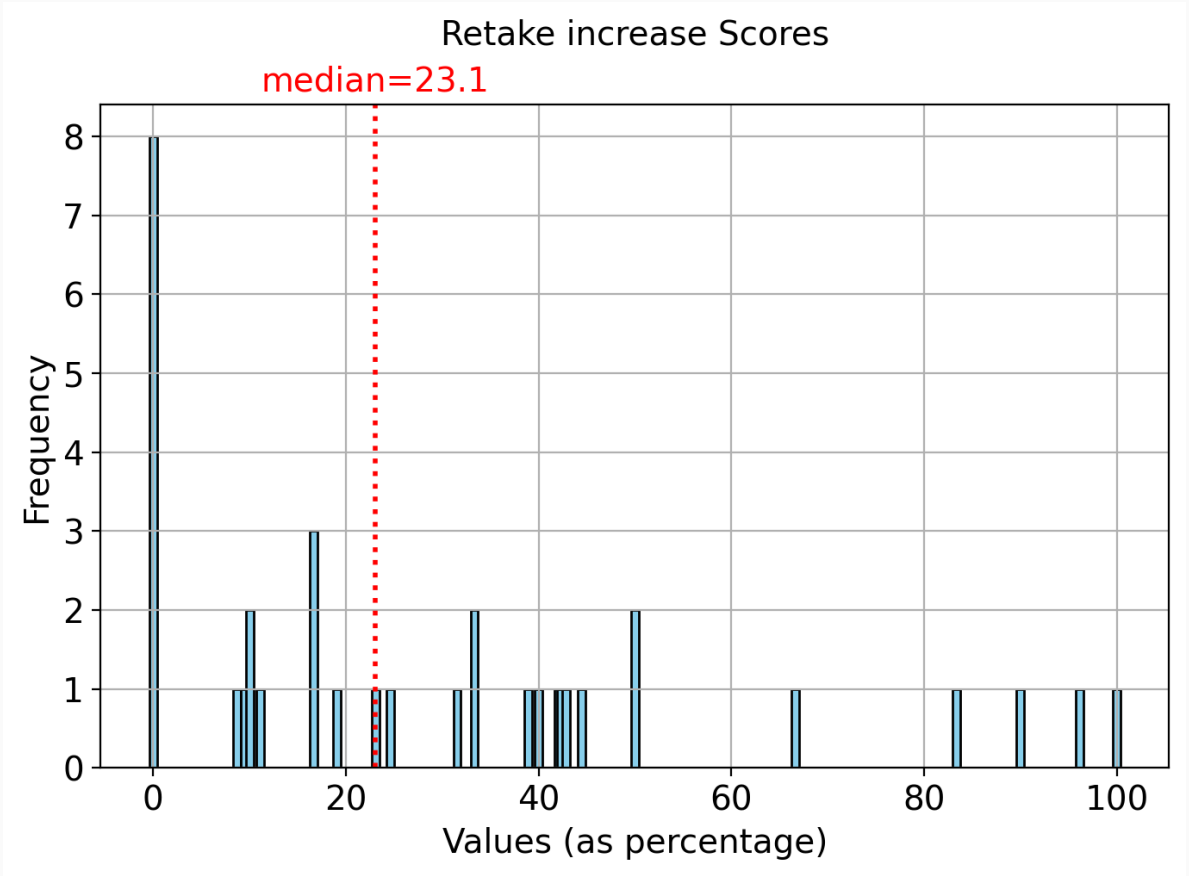
Quiz Retake Results

You improved your scores by quite a bit!



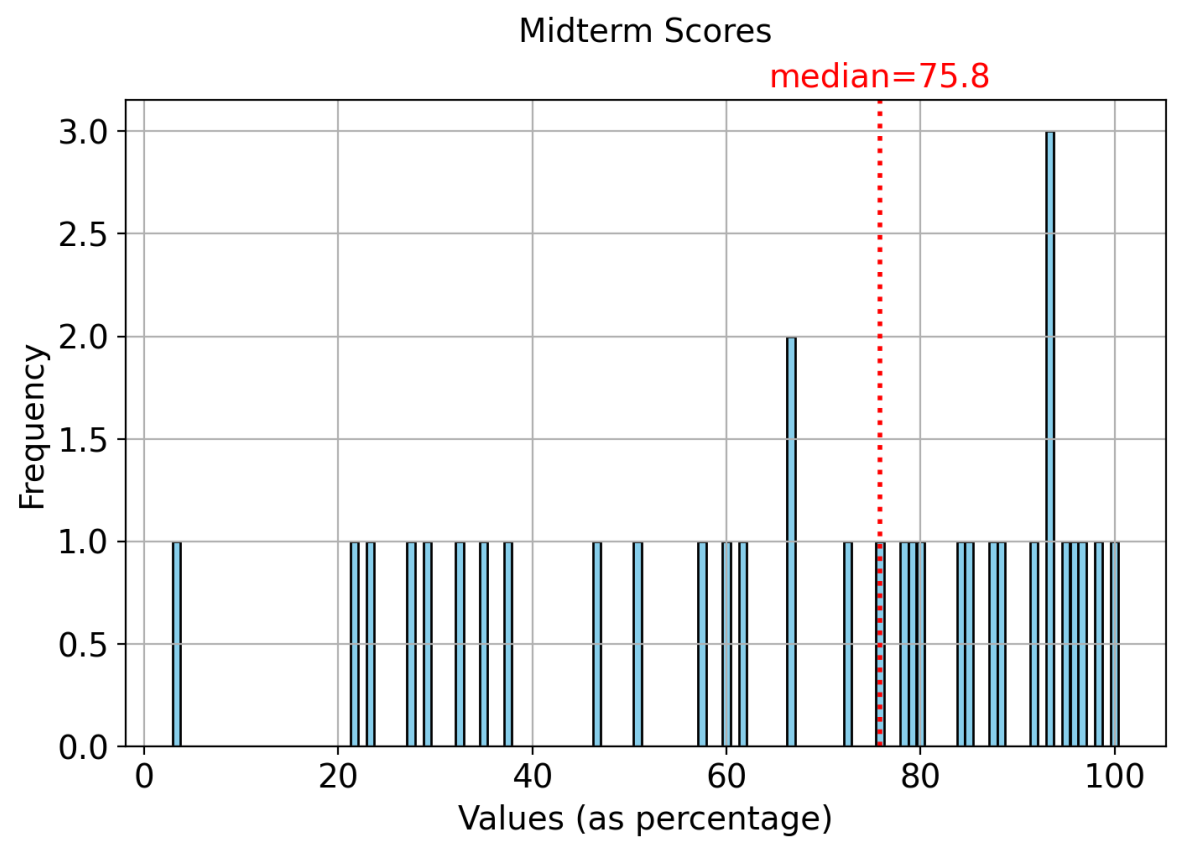
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You improved your scores by quite a bit!



(Apologies to those who had already maxed out their score.)

Midterm Results



What you could do if your midterm score is lower than you'd like?

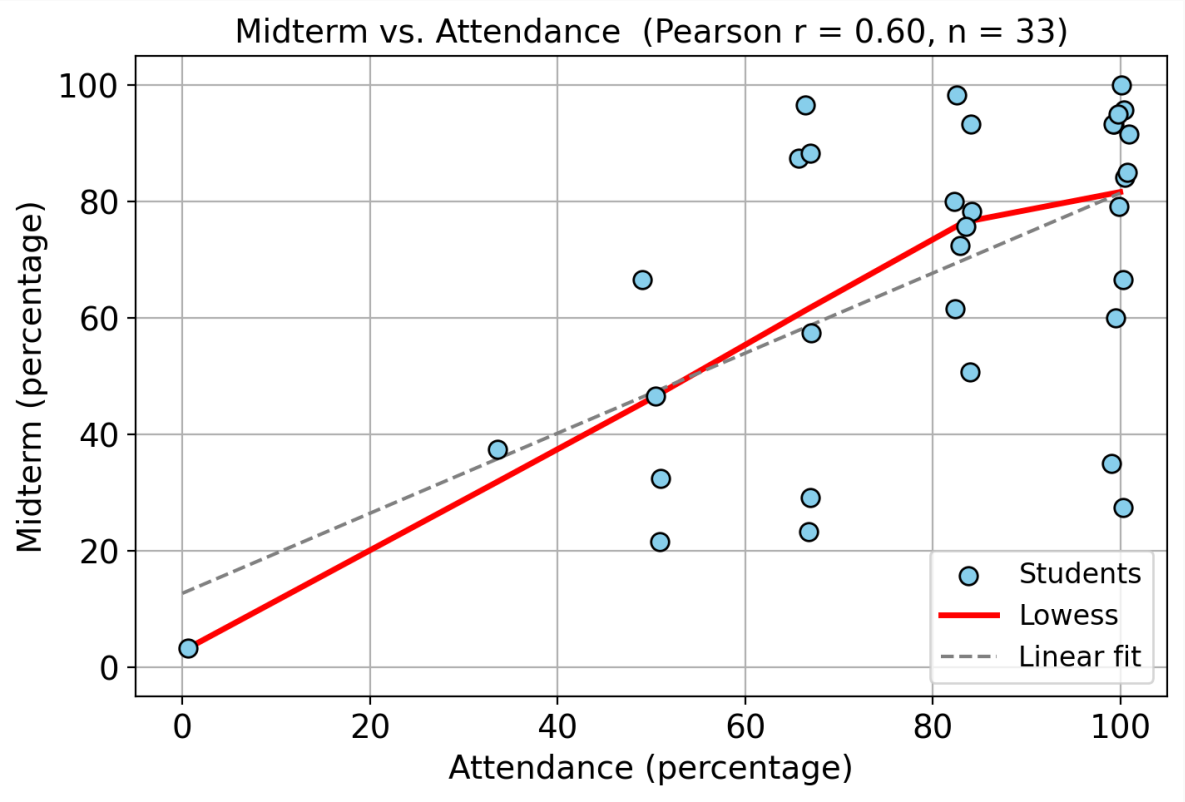
Consider rethinking your preparation strategy...

1. Are you (actively) reading the textbook before class?

What you could do if your midterm score is lower than you'd like?

Consider rethinking your preparation strategy...

1. Are you (actively) reading the textbook before class?
2. Are you attending class?



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Consider rethinking your preparation strategy...

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3. Are you finishing the group exercises after class?

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4. Take advantage of office/tutoring hours.

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1. Are you (actively) reading the textbook before class?
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5. Come chat with me to brainstorm on the details.

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Opportunity. I will offer optional midterm retakes. Details forthcoming.

Study Guide for Quiz 6

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Group exercises. Know how to solve the group exercises from the course meetings on Binomial Coefficients (9/26) and Contradiction and Contraposition (10/3).

Readings.

Sec. 17 (Binomial Coefficients)

- Understand Def 17.1, as used in Examples 17.2-17.4 and 17.6.
- Understand how to apply the computational formula of Theorem 17.12.
- Understand the intuition behind Prop 17.7.

Sec. 20 (Contradiction and Contraposition)

- Understand how to apply Proof Template 11.
- Understand how to apply Proof Template 12, as used to prove Proposition 20.3.

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Sec. 20 (Contradiction and Contraposition)

- Understand how to apply Proof Template 11.
- Understand how to apply Proof Template 12, as used to prove Proposition 20.3.

Extra credit proof review question. I will give 10% extra credit on your midterm if you can answer the review question perfectly.

Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.

For a solution, see Group Exercise #3 from the slide deck on proofs (09/02/26).

Induction

Induction Demo

Good news! I'm going to pay you to learn induction!



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Rule.

- **(Base)** Student $n = 1$ gets a coin.
- **(Step)** If student $n - 1$ gets a coin, then student n gets a coin.

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Student ID	Gets a coin?	Reason

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1	Yes	Base

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1	Yes	Base
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1	Yes	Base
2	Yes	Step applied to student 1
3	Yes	Step applied to student 2

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4	Yes	Step applied to student 3

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⋮	⋮	⋮

Triangular Numbers

Poll

Last class, we proved

$$1 + 2 + 3 + \cdots + n = n(n+1)/2.$$

What is the intuition behind this formula?

Triangular Numbers

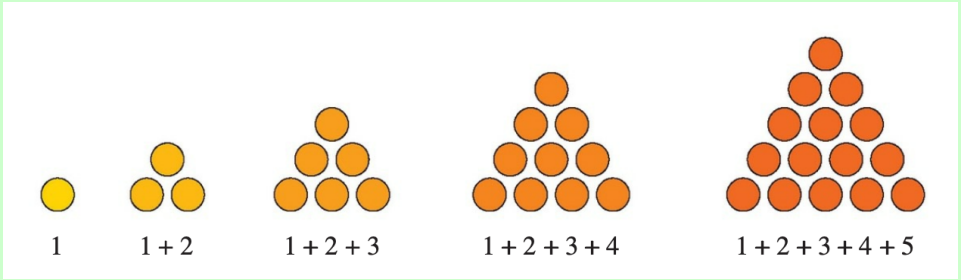
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Hint



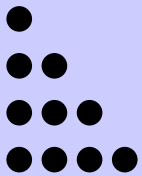
Triangular Numbers: Geometric Intuition

Triangular Numbers

The n-th triangular number is

$$T_n = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

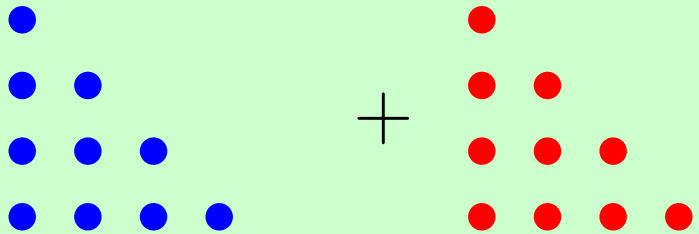
It counts the number of dots you can arrange in an **triangle** like this:



Triangular Numbers: Geometric Intuition

Geometric Intuition: Make a rectangle from two triangles

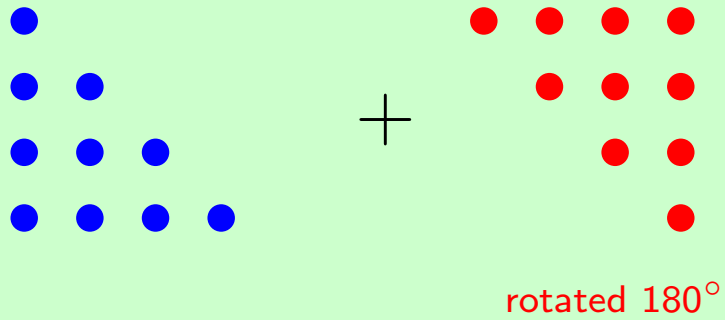
Take **two copies** of the triangle for T_4.



Triangular Numbers: Geometric Intuition

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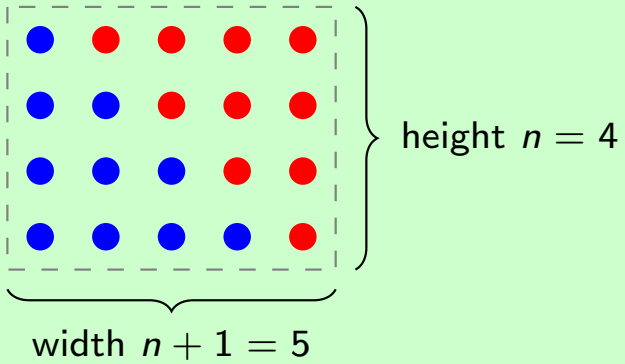
Take **two copies** of the triangle for T_4. Rotate one copy by 180°.



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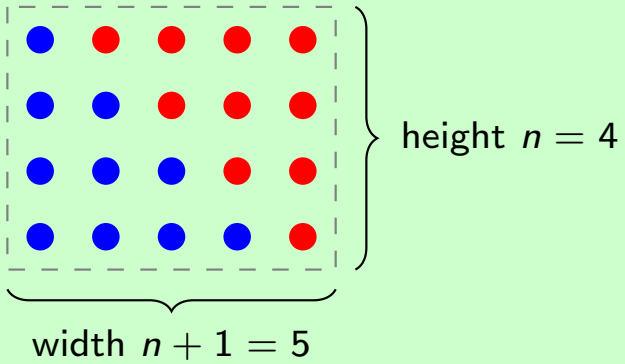
Take **two copies** of the triangle for T_4. Rotate one copy by 180°. Then slide it into place: the two triangles fit together into a **rectangle**.



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Geometric Intuition: Make a rectangle from two triangles

Take **two copies** of the triangle for T_4. Rotate one copy by 180°. Then slide it into place: the two triangles fit together into a **rectangle**.



So the rectangle has **area** $n(n+1)$ dots. But it is made of **two identical triangles**, so the number of dots in each triangle is

$$T_n = \frac{n(n+1)}{2}.$$

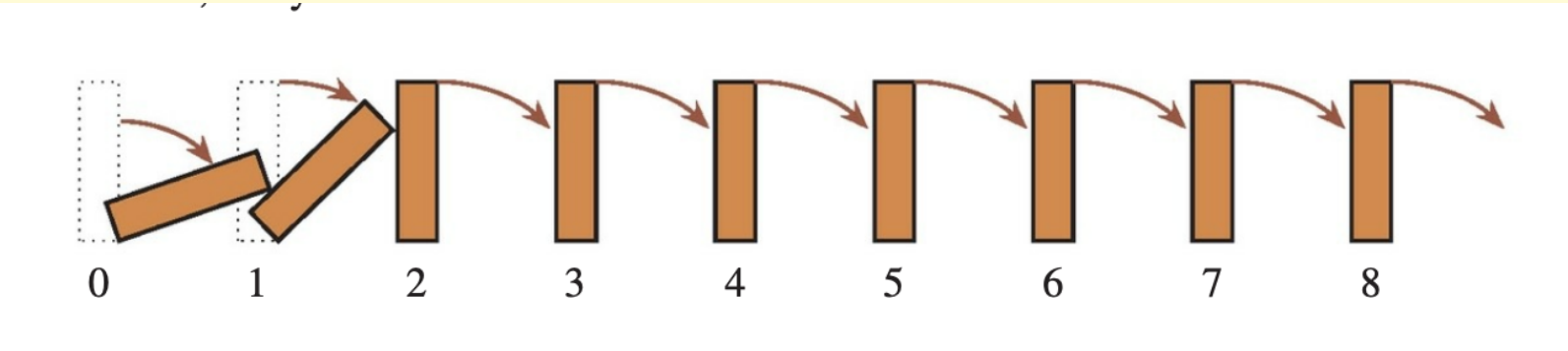
The Induction Strategy to Proving Things

Problem

Prove that the n-th triangular number is $n(n+1)/2$. That is, prove

$$1 + 2 + 3 + \cdots + n = n(n+1)/2.$$

Induction Strategy



Base. Show the equation holds for some starting point (usually 0 or 1).
Induction step. Show that if the equation holds for natural number $n = k$, then it must hold for natural number $n = k + 1$.

The strategy in the yellow box lets us conclude the equation holds for all n.

Proof By Induction: Example

Problem (Induction)

Prove that the n-th triangular number is $n(n+1)/2$. That is, prove

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}. \tag{1}$$

Base. We show Eq. (1) holds for $n = 1$. In that case, Eq. (1) says

$$1 = \frac{1(1+1)}{2}$$

which is true.

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Induction step. We assume Eq. (1) holds for $n = k$, where $k \geq 1$, which was the base case. (This is called the induction hypothesis.) We show Eq. (1) holds for $n = k + 1$. That is, we must show

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}. \tag{2}$$

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We have

$$\text{LHS of Eq. (2)} = \frac{k(k+1)}{2} + (k+1) \qquad \text{(by induction hypothesis)}$$

$$= \text{RHS of Eq. (2)} \qquad \text{(after some algebra)}$$

Induction and Strong Induction

What we've been using so far is classical induction.

Scheinerman Theorem 22.2: Induction

Let A be a set of natural numbers. If

- $0 \in A$, and
- $\forall k \in \mathbb{N}, k \in A \implies k+1 \in A$

then $A = \mathbb{N}$.

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There is also a strong version.

Scheinerman Theorem 22.9: Strong Induction

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Remark. If the result you wish to prove doesn't cover all natural numbers – say it covers just the positive integers – the basis step may begin at a value other than 0.

Strong Induction

When to Use Strong Induction

Sometimes there are scenarios where the induction implication goes naturally not from n to $n+1$, but:

- From some smaller number to $n+1$, or
- From several smaller numbers

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Example: Every positive integer has a prime factorization

- Suppose every number smaller than n has a prime factorization.
- The number n must either be prime (in which case we are done) or composite.
- If n is composite, then $n = ab$ for smaller number $a, b < n$. But a, b have their own factorizations by the induction hypothesis. These can be put together to make a prime factorization of n.

Uses of induction in computer science: Examples

1. **Proving correctness of recursive algorithms:** E.g. we can use induction to show `factorial`$(n) = n!$.

```
def factorial(n):
    """Compute n! recursively"""
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)
```

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2. **Analyzing algorithm complexity:** To determine the complexity – e.g. $\mathcal{O}(n \log n)$ – of recursive algorithms (like mergesort, quicksort, binary search), we often use induction on n.

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3. **Proving properties of data structures:** Many structures (like linked lists, binary trees, heaps, etc.) are defined inductively.

Group exercises

Random group assignments

Ashley, Isabelle: 3
Bailey, Tanna: 13
Blaisdell, Olyver: 13
Brown, Ashton: 5
Brown, Jonathan: 1
Cardenas, Dominic: 11
Cutler, George: 9
Fabik, Roland: 3
Fellows, Audrey: 8
Fitzgerald, Ryan: 5
Fuller, Ryan: 6
Griffen, Clara: 10
Grutsch, Tanner: 9
Hager, Nathan: 7
Horner, Jayden: 12
Johnson, Jaret: 2
Kintzel, Samantha: 4
Lameres, Alexis: 4
Lee, Holden: 1
Mayala, Reece: 5
McLean, Connor: 7
Mcbride, Daniel: 3
Miller, Maverick: 10
Moss, Emily: 6
Mousseau, Matthew: 12
Neuman, Evan: 7
Putnam, John: 2
Ramos, Vydel: 8
Rathbun, Jedidiah: 11
Roberts, Matthew: 10
Roller, Nathan: 1
Ryan, Otis: 11
Shevtsov, Timothy: 4
Sinclair, Aidan: 8
Stensland, Emma: 2
Thiessen, Logan: 9
Thompson, Jaiden: 6
Vath, Ethan: 12

Group exercises: Induction

1. Show by induction that $n^2 - n$ is even for any natural number n (that is, for $n = 0, 1, 2, \ldots$).
2. Show by induction that $2^n < n!$ for all $n \geq 4$.
3. Show by induction that $f_0 + \cdots + f_n = f_{n+2} - 1$ in the Fibonacci sequence.
4. Show by induction that $f_n < 2^n$ in the Fibonacci sequence.

Definition

The **Fibonacci sequence** is the sequence given by the following recursive rules:

$$f_0 = 0, \qquad f_1 = 1, \qquad f_{n+2} = f_n + f_{n+1}$$

The sequence is: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$

Solutions to group exercises

Exercise 1

Show by induction that $n^2 - n$ is even for any natural number n (that is, for $n = 0, 1, 2, \ldots$).

Solution (by Induction)

Induction base. Take $n = 0$. Then $n^2 - n = 0^2 - 0 = 0$, which is even. So $n^2 - n$ is even when $n = 0$.

Induction step. Now we assume the proposition holds at index $k \geq 0$ (this is called the induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $k^2 - k$ is even, and we need to show that $(k+1)^2 - (k+1)$ is even. We have

$$\begin{aligned}(k+1)^2 - (k+1) &= (k^2 + 2k + 1) - (k+1) \\ &= k^2 + k \\ &= \underbrace{(k^2 - k)}_{\text{even by induct. hypoth.}} + \underbrace{(2k)}_{\text{even by def. even}} \qquad \text{(subtract, add } n\text{)}\end{aligned}$$

And the sum of two even numbers is even.

Hence, $k^2 - k$ is even $\implies (k+1)^2 - (k+1)$ is even.

Exercise 2

Show by induction that $2^n < n!$ for all $n \geq 4$.

Solution

Induction base. Take $n = 4$. Then $2^n = 2^4 = 16$ and $n! = 4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. So $2^n < n!$ when $n = 4$.

Induction step. Now we assume the proposition holds at index $k \geq 4$ (this is called the induction hypothesis), and we need to show it holds at index $k+1$. That is, we assume $2^k < k!$ (this is called the induction hypothesis), and we need to show $2^{k+1} < (k+1)!$ We have

$$
\begin{aligned}
2^{k+1} &= 2 \cdot 2^k \\
&< 2 \cdot k! && \text{(by the induction hypothesis)} \\
&< (k+1) \cdot k! && \text{(since } k \geq 4\text{)} \\
&= (k+1)! && \text{(by def. factorial)}
\end{aligned}
$$

Hence, $2^k < k! \implies 2^{k+1} < (k+1)!$

Exercise 3

Show by induction that $f_0 + \cdots + f_n = f_{n+2} - 1$ in the Fibonacci sequence.

Solution

Induction base. Take $n = 0$. Then $f_0 + \cdots + f_n = f_0 = 0$ and $f_{n+2} - 1 = f_2 - 1 = 1 - 1 = 0$. So $f_0 + \cdots + f_n = f_{n+2} - 1$ when $n = 0$.

Induction step. Now we assume the proposition holds at index $k \geq 0$ (this is called the induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $f_0 + \cdots + f_k = f_{k+2} - 1$, and we need to show that $f_0 + \cdots + f_{k+1} = f_{k+3} - 1$. We have

$$\begin{aligned} f_0 + \cdots + f_{k+1} &= (f_0 + \cdots + f_k) + f_{k+1} \\ &= (f_{k+2} - 1) + f_{k+1} && \text{(by the induction hypothesis)} \\ &= (f_{k+1} + f_{k+2}) - 1 \\ &= f_{k+3} - 1 && \text{(by def. Fibonacci number)} \end{aligned}$$

Hence, if the proposition holds at index k, it holds at index $k + 1$.

Exercise 4

Show by induction that $f_n < 2^n$ in the Fibonacci sequence.

Solution

Induction base. Take $n = 0$. Then $f_n = f_0 = 0$ and $2^n = 2^0 = 1$. So $f_n < 2^n$ when $n = 0$.

Induction step. Here we use strong induction. We assume the proposition holds at indices $0, 1, \ldots, k$ for $k \geq 0$ (this is the (strong) induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $f_j < 2^j$ for $j = 0, 1, \ldots, k$, and we need to show that $f_{k+1} < 2^{k+1}$. We have

$$
\begin{aligned}
f_{k+1} &= f_k + f_{k-1} && \text{(by def. Fibonacci number)} \\
&< 2^k + 2^{k-1} && \text{(by the (strong) induction hypothesis)} \\
&< 2^k + 2^k \\
&= 2 \cdot 2^k \\
&= 2^{k+1}.
\end{aligned}
$$

Hence, $f_k < 2^k \implies f_{k+1} < 2^{k+1}$.